

$$\begin{array}{lll} \bar{n}_1(0) = 0 & V_1 = 1 & S_1 = 10 \\ \bar{n}_2(0) = 0 & V_2 = 0.5 & S_2 = 15 \\ \bar{n}_3(0) = 0 & V_3 = 0.5 & S_3 = 30 \end{array}$$

$$R_1(1) = (1 + \bar{n}_1(0)) S_1 = 10$$

$$R_2(1) = (1 + \bar{n}_2(0)) S_2 = 15$$

$$R_3(1) = (1 + \bar{n}_3(0)) S_3 = 30$$

$$\begin{aligned} R_0(1) &= V_1 R_1(1) + V_2 R_2(1) + V_3 R_3(1) \\ &= 1 * 10 + 0.5 * 15 + 0.5 * 30 = 32.5 \end{aligned}$$

$$X_0(1) = 1 / R_0(1) = 1 / 32.5$$

$$X_1(1) = V_1 X_0(1) = 10 / 32.5 = 0.0308$$

$$X_2(1) = V_2 X_0(1) = 0.5 / 32.5 = 0.0154$$

$$X_3(1) = V_3 X_0(1) = 0.5 / 32.5 = 1/65$$

$$\begin{aligned} \bar{n}_1(1) &= R_1(1) X_1(1) = 10 * 0.0308 = \\ &0.308 \end{aligned}$$

$$\bar{n}_2(1) = R_2(1) X_2(1) = 15 * 0.0154 = 0.231$$

$$\bar{n}_3(1) = R_3(1) X_3(1) = 30 * 0.0154 = 0.462$$

$$\begin{array}{lll} \bar{n}_1(1) = 0.3077 & V_1 = 1 & S_1 = 10 \\ \bar{n}_2(1) = 0.2310 & V_2 = 0.5 & S_2 = 15 \\ \bar{n}_3(1) = 0.4620 & V_3 = 0.5 & S_3 = 30 \end{array}$$

$$R_1(2) = (1 + \bar{n}_1(1)) S_1 = (1 + 0.3077) * 10 = 13.007$$

$$R_2(2) = (1 + \bar{n}_2(1)) S_2 = (1 + 0.2310) * 15 = 18.46$$

$$R_3(2) = (1 + \bar{n}_3(1)) * S_3 = (1 + 0.462) * 30 = 43.8$$

$$\begin{aligned} R_0(2) &= V_1 R_1(2) + V_2 R_2(2) + V_3 R_3(2) \\ &= 1 * 13.007 + 0.5 * 18.46 + 0.5 * 43.8 = 44.23 \end{aligned}$$

You can continue to find:

$$X_0(2), X_1(2), X_2(2), X_3(2)$$

$$\bar{n}_1(2), \bar{n}_2(2), \bar{n}_3(2)$$

Scratch: